DALAS Project
Drug Repurposing

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1 Introduction

1.1 Problem Statement

Clearly define the problem your project addresses. Explain its relevance in the scientific or practical context.

The discovery of new drugs is, although crucial, an extremely expensive and time-consuming process. Many candidate drugs ultimately turn out to be not only ineffective, but also potentially toxic. It is thus of interest to repurpose already approved drugs for diseases they have not yet been tested against. To prioritize potential clinical trials, in addition to experts' knowledge, it's important to be able to predict the likelihood of success for such studies. This project aims to explore the data and develop a potential prediction pipeline for this purpose. We acknowledge that industrial and academic drug development employ far more comprehensive methods, this project, by contrast, focuses on a preliminary, data-driven approach and is exploratory in nature.

1.2 Motivation and Objectives

Explain why this problem is important. State your main objectives and hypotheses.

The ability to predict the success of drug repurposing has academic and clinical value, it helps reducing the experimental cost and accelerating the discovery of effective treatments. The main objectives of this project are to:

- Explore public datasets to identify some relevant features
- Develop a preliminary machine learning pipeline for success prediction
- Evaluate the model performance
- Identify biases and potential ways for improvement

We decided to specify a group of diverse but closely related diseases - immune system disorders and drugs that are approved for at least one member of this group.

2 Data

2.1 Data Sources

Describe where the data comes from, collection methods, and aggregation process.

Among the vast amount of medical, pharmacological and chemical publicly available data, we focused our attention on those being able to provide us

Table 1: Description of data sources.

Source	Access Method	Data Retrieved
MeSH RDF	SPARQL endpoint	Disease IDs and ontology
ChEMBL	REST API (Python client with query-style filters)	Drug chemical properties, targets and indications
UniProt	REST API (Python client)	Map target and pathway IDs
Open Targets	GraphQL API	Disease targets
ClinicalTrials.gov	REST API	Trial results and metadata

2.2 Data Quality and Preprocessing

Discuss missing values, outliers, normalization, feature encoding, or transformations.

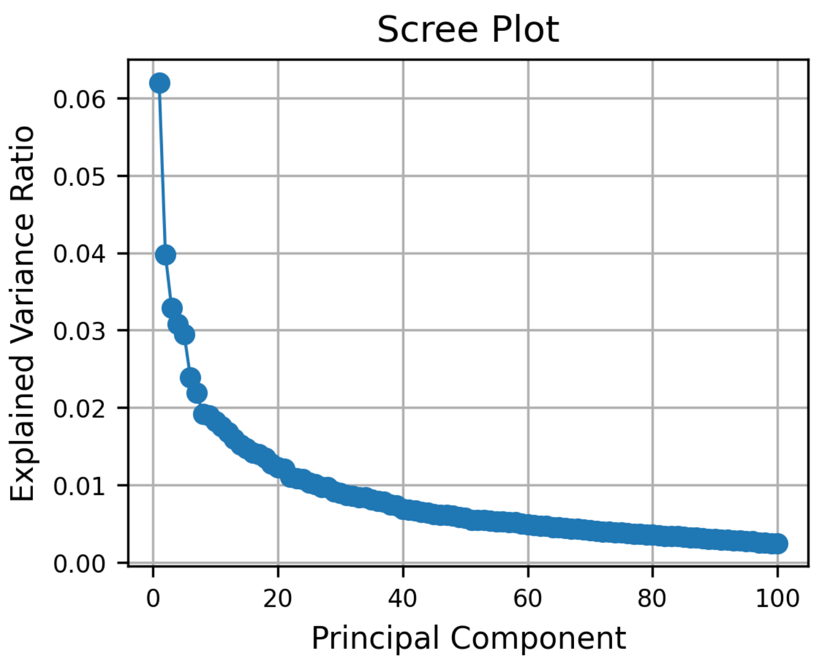

Figure 1: Scree plot of PCA on merged data.

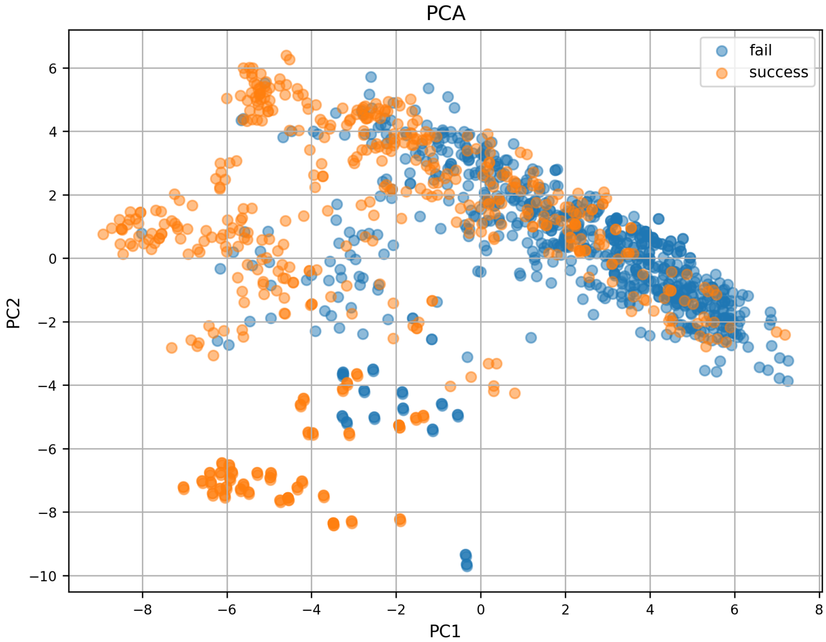

Figure 2: Scree plot of PCA on merged data.

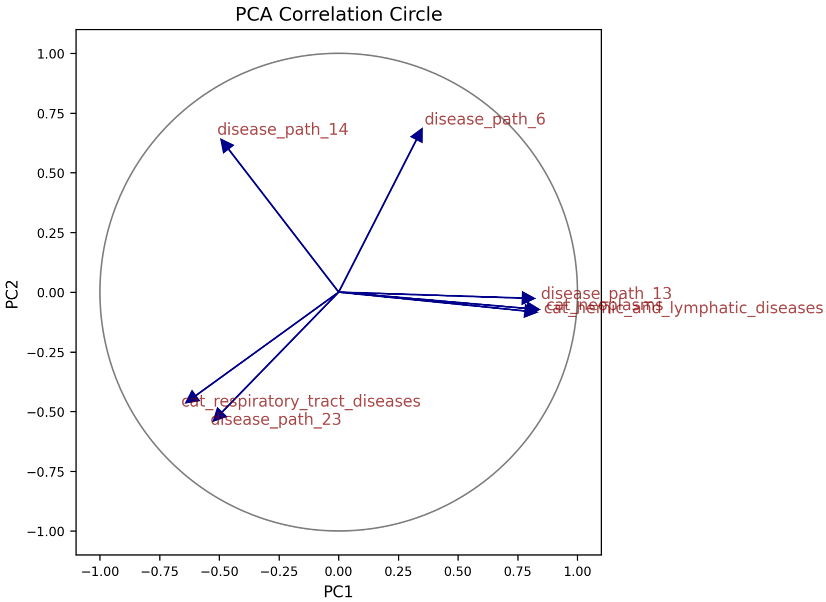

Figure 3: Scree plot of PCA on merged data.

2.3 Feature Description

Provide a table or list explaining each feature and its meaning. Optionally, include relationships between features.

Table 2: Summary of datasets and merged data.

Dataset	Rows	Columns
Drugs	742	181
Diseases	167	79
Trials	7023	11
Indications	3236	11
Name embeddings	489939	5
Molecular fingerprints	1547	101
Merged data	1298	255
Merged data success counts	False: 741, True: 557	

Table 3: Description of features.

Feature	Type	Description
Drug		
drug_id	str	Unique ChEMBL identifier
chemical_probe	bool	Well-characterized and selective, typically used in biological research
dosed_ingredient	bool	Active ingredient that has been dosed in humans
inorganic_flag	bool	Inorganic molecule
natural_product	bool	Derived from a natural source (not fully synthetic)
oral	bool	Known to be administered orally
orphan	bool	Intended for use against a rare condition
parenteral	bool	Known to be administered parenterally
prodrug	bool	Prodrug, activated after administration
therapeutic_flag	bool	Has a therapeutic application, opposed to e.g. imaging
topical	bool	Known to be administered topically
veterinary	bool	Has veterinary application as well
chirality	cat	achiral,single enantiomer, mixture,nan
alogp	float	Octanol-water partition coefficient, lipophilicity indicator
aromatic_rings	int	Number of aromatic rings
full_mwt	float	Molecular weight of the full compound including any salts
hba	float	Number of hydrogen bond acceptors
hbd	float	Number of hydrogen bond donors
heavy_atoms	int	Number of heavy (non-hydrogen) atoms
mw_freebase	float	Molecular weight of parent compound
np_likeness_score	float	Natural product likeness score [Ertl et al., 2008]
num_ro5_violations	int	Violations of Lipinski's rule-of-five, using HBA and HBD definitions
psa	float	Polar surface area
qed_weighted	float	Weighted quantitative estimate of drug likeness [Bickerton et al., 2012]
ro3_pass	int	Passes the rule-of-three (mw $<$ 300, logP $<$ 3 etc)
rtb	int	Number of rotatable bonds
FP_#	float	First PCs in Morgan molecular fingerprint space
drug_path_#	float	First PCs in TF-IDF space derived from pathway lists
Disease		
disease_id	str	Unique EFO identifier
ontology_depth	int	Depth in disease classification tree
nb_descendants	int	Number of descendants in disease classifications tree
nb_indications	int	Number of drug indications
cat_#	bool	Ontological category other than immune system disorders
disease_path_#	float	First PCs in TF-IDF space derived from pathway lists
Drug–Disease		
success	bool	Consensus success or failure
path_similarity	float	Jaccard similarity of pathway lists
name_similarity	float	Cosine similarity of name embeddings in PubMedBERT (later excluded due to data leak)

```
  HeteroData(
drug={ x=[742, 127] },
disease={ x=[167, 25] },
pathway={ x=[1509, 1] },
(drug, interacts, pathway)={ edge_index=[2, 31266] },
```

```
  (disease, associated_with, pathway)={ edge_index=[2, 13608] },
  (drug, tryed_against, disease)={
    edge_index=[2, 1298],
    edge_label=[1298],
  }
)
```

3 Methods

3.1 Model Overview

Briefly describe the models used. Include general principles (e.g., linear models, random forests, neural networks).

3.2 Implementation Details

Mention important parameters, software/libraries used, and cross-validation setup.

4 Results

4.1 Model Comparison

Compare models quantitatively (accuracy, RMSE, AUC, etc.) and qualitatively (strengths/weaknesses).

Table 4: Model comparison summary.

model	accuracy	precision	recall	f1_score	roc_auc	params	mean_test_score	std_test_score
Support Vector Machine	0.776923	0.825000	0.600000	0.694737	NaN	C = 1, degree = 3, gamma = auto, kernel = poly	0.867400	0.076985
Random Forest	0.753846	0.828571	0.527273	0.644444	0.829818	max_depth = None, min_samples_leaf = 1, min_samples_split = 5, n_estimators = 100	0.875187	0.060380
Gradient Boost	0.753846	0.848485	0.509091	0.636364	0.851394	learning_rate = 0.01, max_depth = 5, n_estimators = 100, subsample = 0.8	0.873543	0.048047
Logistic Regression	0.715385	0.736842	0.509091	0.602151	0.797333	C = 0.01, l1_ratio = None, penalty = l2	0.833953	0.074911
k-Nearest Neighbors	0.676923	0.675676	0.454545	0.543478	0.716848	n_neighbors = 30, p = 1, weights = distance	0.850727	0.082253

4.2 Feature Analysis

Analyze the impact of key features on model performance (e.g., feature importance, correlation).

4.3 Failure and Success Cases

Highlight scenarios where the models perform well or poorly. Include illustrative figures or tables.

5 Discussion

Interpret the results. Relate findings to your original objectives. Discuss limitations and possible improvements.

6 Conclusion

Summarize your work, key findings, and potential future directions.

References

[Bickerton et al., 2012] Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S., and Hopkins, A. L. (2012). Quantifying the chemical beauty of drugs. *Nat Chem*, 4(2):90–98.

[Ertl et al., 2008] Ertl, P., Roggo, S., and Schuffenhauer, A. (2008). Natural product-likeness score and its application for prioritization of compound libraries. *J. Chem. Inf. Model.*, 48(1):68–74. Epub 2007 Nov 23.